

PROJECT REPORT

Auto-Ticket Classification using Flow Designer

ServiceNow School IT Helpdesk Automation

Team Member	Primary Responsibility
Aditya A	Sprint 1 - Requirement Analysis & Planning
Akash N A	Sprint 2 - Backend Development & Schema Configuration
Radha Krishnan P	Sprint 3 - Flow Designer Automation
Ferdin Joe A	Sprint 4 - Testing & Validation
Simeon P	Sprint 5 - Deployment & Packaging

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1. INTRODUCTION

1.1 Project Overview

Auto-Ticket Classification using Flow Designer is a ServiceNow-based school IT helpdesk automation project. It provides a dedicated School IT Ticket record for support requests and moves the initial classification step into an automated workflow.

The implementation includes the custom table `u_school_it_ticket`, the School IT Support application/module, core ticket fields, dependent Category and Subcategory choices, the Auto Classify School IT Tickets Flow Designer workflow, requester email notification, access-control configuration, and Update Set based change tracking.

1.2 Purpose

The purpose of the project is to simplify the student ticket-submission experience and reduce repetitive manual classification work for the helpdesk. A student can provide the Caller and Short Description, while the workflow identifies supported issue types and sets the corresponding Category and Subcategory.

The project is focused on a practical school IT support scenario in which common requests such as Wi-Fi, projector, password/login, and slow or hanging computer issues need a consistent initial classification.

2. IDEATION PHASE

2.1 Problem Statement

Students may not know which technical category or subcategory matches an IT issue. Helpdesk technicians may therefore need to spend additional time reviewing and classifying incoming requests before processing them.

Student perspective: I am a student using the school IT helpdesk. I am trying to submit an IT support request quickly and correctly, but I may not know which category or subcategory to select because different IT issues require different classifications.

Technician perspective: I am an IT helpdesk technician. I am trying to process incoming support requests efficiently, but I have to review and classify tickets because students may not select the appropriate category.

2.2 Empathy Map Canvas

Area	Student	IT Helpdesk Technician
Think & Feel	Wants quick reporting; may be unsure about technical categories; wants correct routing.	Wants correctly classified tickets; wants less repetitive triage; wants to start troubleshooting quickly.
Hear	Category and submission instructions; helpdesk guidance.	Student complaints, requests for quick resolution, ticket-status questions.
See	School IT Ticket form, categories, subcategories and technical terminology.	Incoming tickets, issue descriptions, categories, subcategories and unclassified requests.
Say & Do	Reports Wi-Fi, projector, password or performance issues; enters Caller and Short Description; submits.	Reviews tickets, identifies issues, checks classification and processes requests.
Pain	Category confusion, wrong selection, extra effort and possible delay.	Manual classification, repetitive triage, inconsistent categorization and delayed troubleshooting.
Gain	Simple submission, automatic classification, acknowledgment and clearer routing.	Consistent classification, reduced triage and organized ticket processing.

2.3 Brainstorming

The team considered several approaches: manual category selection, keyword-based classification, rule-based classification, AI-based classification, a simple ticket form, dependent choices, email acknowledgment, and automatic assignment.

The implemented prototype uses rule and keyword-based Flow Designer logic. This choice keeps the requester-facing interaction simple and fits the documented ServiceNow implementation.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

Stage	Customer Activity	Interaction	Goal	Pain Point	Opportunity
Entice	Student notices an IT problem.	Issue occurs in a classroom, library, lab or office.	Recognize that support is needed.	Uncertainty about the correct helpdesk path.	Provide a clear support entry point.
Enter	Student opens School IT Tickets.	ServiceNow application/module.	Reach the ticket form quickly.	Technical interface may be unfamiliar.	Keep navigation simple.
Engage	Student enters Caller and Short Description.	School IT Ticket form.	Submit the issue with minimum effort.	Technical classification can be confusing.	Move classification into automation.
Exit	Flow processes the record.	Flow Designer updates Category/Subcategory.	Know the ticket has been structured.	Vague or unsupported wording may need review.	Expand rules and add fallback handling.
Extend	Requester receives acknowledgment; helpdesk processes the record.	Email and ticket list.	Know that the request was received.	Final resolution still depends on the helpdesk.	Add future status and assignment notifications.

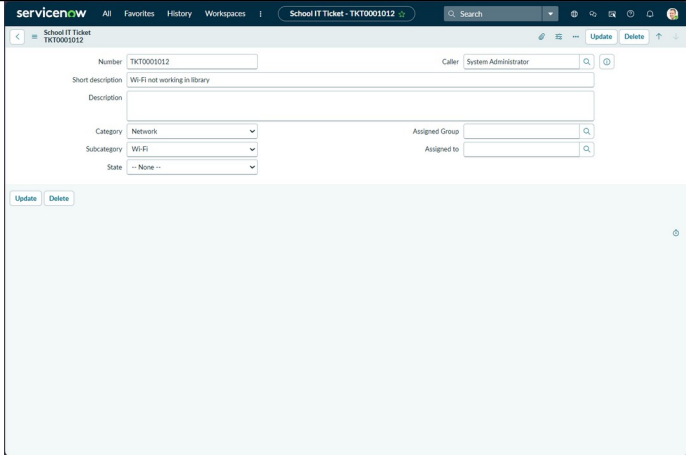


Figure 1: School IT Ticket form used during the ticket-submission journey.

3.2 Solution Requirement

Type	Requirement	Description
Functional	Ticket Submission	Create a School IT Ticket using Caller and Short Description.
Functional	Automatic Classification	Evaluate supported keywords and set Category and Subcategory.
Functional	Dependent Choice	Keep Subcategory dependent on Category.
Functional	Notification	Send an acknowledgment email to the Caller.
Functional	Ticket Processing	Store and display ticket records for the helpdesk.
Functional	Change Tracking	Capture customizations in the Project Update Set.
Non-functional	Usability	Keep the requester-facing input simple.
Non-functional	Security	Use ServiceNow ACL and role controls.
Non-functional	Reliability	Use consistent rules for supported keywords.
Non-functional	Availability	Provide the workflow through the ServiceNow platform.
Non-functional	Scalability	Allow the same flow to process repeated ticket records.

3.3 Data Flow Diagram

The data flow begins with the student entering ticket information. The custom record is stored in ServiceNow, Flow Designer evaluates the Short Description, and the resulting classification and notification support the helpdesk process.

Data Flow Diagram - Auto-Ticket Classification

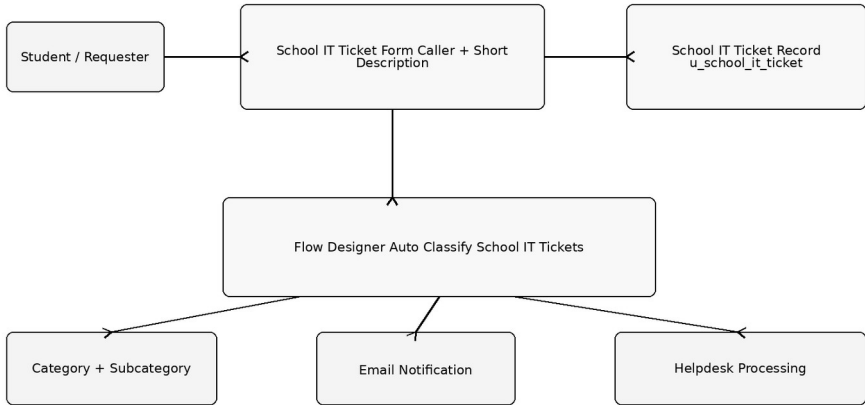


Figure 2: Data Flow Diagram for the automatic ticket-classification process.

3.4 Technology Stack

Layer / Component	Implementation
User Interface	ServiceNow Next Experience - School IT Tickets
Custom Data Model	School IT Ticket table - u_school_it_ticket
Automation	ServiceNow Flow Designer - Auto Classify School IT Tickets
Classification	Rule/keyword-based conditions
Notification	ServiceNow Email / SMTP
Security	ServiceNow ACLs and Roles
Change Management	ServiceNow Update Sets
External API / ML Model	Not used in the implemented workflow
Separate Cloud Database	Not used; ticket data is stored in the ServiceNow platform

Technical Architecture - School IT Ticket Automation

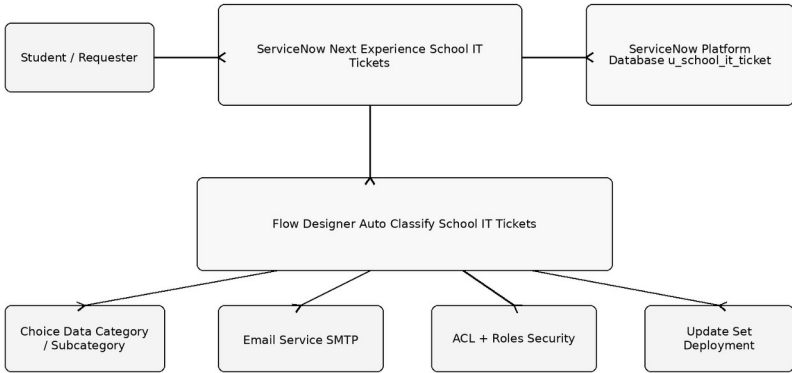


Figure 3: Technical architecture of the ServiceNow implementation.

4. PROJECT DESIGN

4.1 Problem Solution Fit

The solution fits the identified problem by shifting supported classification work from the requester to the ServiceNow workflow. The student supplies basic information that is natural to provide, while the workflow applies the internal classification rules.

For the implemented prototype, Wi-Fi or Network language maps to Network / Wi-Fi, projector-related language maps to Hardware / Projector, password or login language maps to Access / Forgot Password, and slow or hanging computer language maps to Performance / Slow Computer.

4.2 Proposed Solution

- Create the School IT Ticket custom table and School IT Support application/module.
- Capture Caller and Short Description from the requester.
- Use dependent Category and Subcategory choices.
- Trigger the classification flow when a new ticket is created with an empty Category.
- Apply branches for network, hardware, access and performance keywords.
- Send an acknowledgment email to the Caller.
- Use an Update Set to track and export the configuration for deployment.

4.3 Solution Architecture

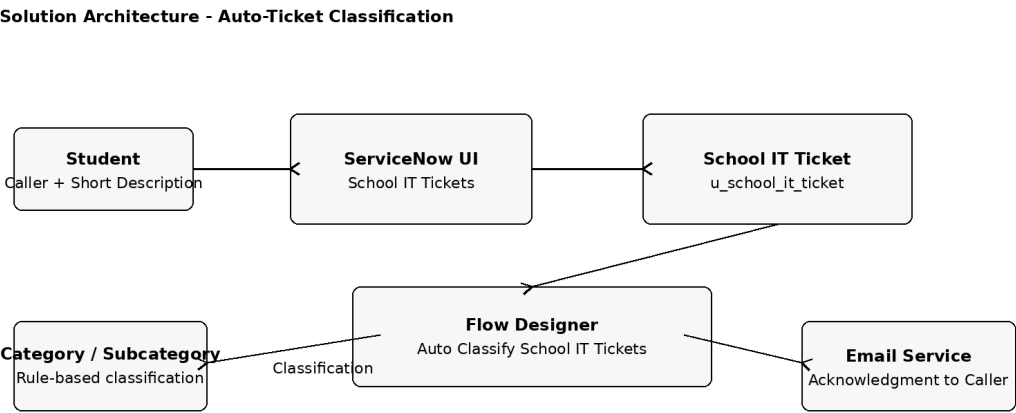


Figure 4: Solution architecture connecting the ServiceNow user interface, ticket data, Flow Designer and notification.

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Sprint	Schedule	Member	Milestone	Story Points
Sprint-1	28 Sep 2026 - 28 Sep 2026	Aditya A	Requirement Analysis & Planning	5
Sprint-2	28 Sep 2026 - 28 Sep 2026	Akash N A	Backend Development & Schema Configuration	11
Sprint-3	29 Sep 2026 - 29 Sep 2026	Radha Krishnan P	Flow Designer Automation	11
Sprint-4	30 Sep 2026 - 30 Sep 2026	Ferdin Joe A	Testing & Validation	5
Sprint-5	01 Oct 2026 - 01 Oct 2026	Simeon P	Deployment & Packaging	5
Total	-	Project Team	All five milestones	37

The five one-day sprints were assigned to the team members specified for this project. The 37 story-point total is a planning estimate mapped to the documented project work items.

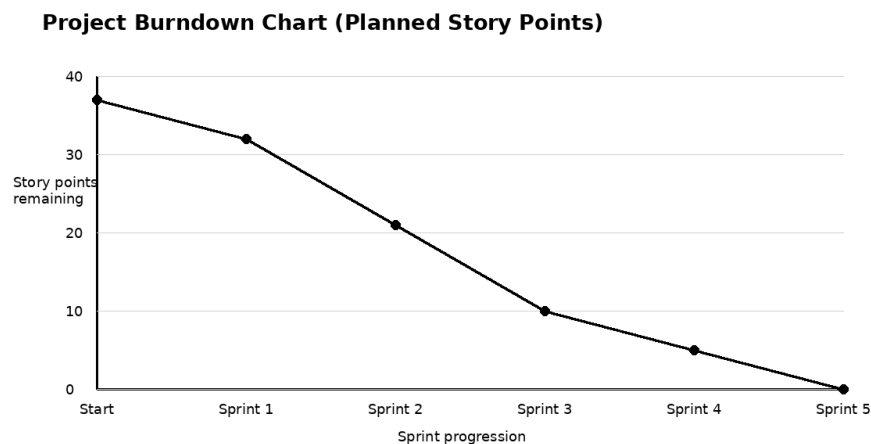


Figure 5: Planned burndown chart based on the 37-point project plan.

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

The supplied testing format requires test scenarios, steps, expected and actual results, and pass/fail status. The implemented project provides functional evidence but does not include measured response-time, concurrent API, or file-upload benchmark results. Those unsupported measures are therefore recorded as Not Tested or N/A.

Test ID	Scenario	Expected Result	Actual Result	Status
FT-01	Create School IT Ticket	Record created with Caller and Short Description.	Created School IT Ticket records are visible in the ticket list.	Pass
FT-02	Automatic Wi-Fi Classification	Category = Network and Subcategory = Wi-Fi.	The demonstrated ticket shows Network and Wi-Fi.	Pass
FT-03	Projector / Hardware Classification	Projector request maps to Hardware / Projector.	A projector request is shown as Hardware in the ticket list.	Pass
FT-04	Email Acknowledgment	Acknowledgment email is generated for the Caller.	Email log and preview show the acknowledgment message.	Pass
PT-01	Response Time Measurement	Execution time is measured and recorded.	No timing measurement was supplied.	Not Tested
PT-02	Concurrent API Load	External API remains responsive.	N/A; no external classification API is used.	N/A
PT-03	File Upload Load	Multiple files process without failure.	N/A; file upload is not part of the workflow.	N/A

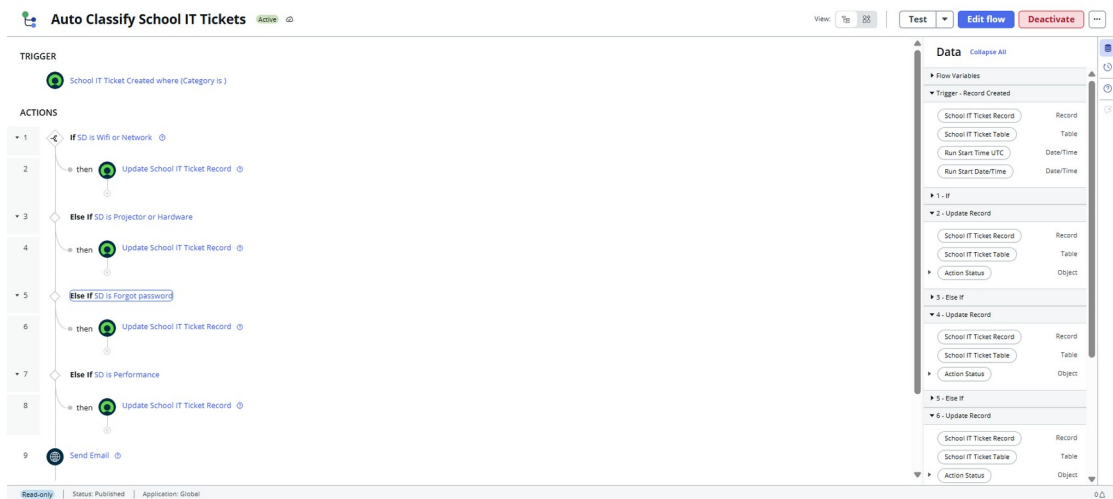


Figure 6: Published Flow Designer workflow used for the functional classification process.

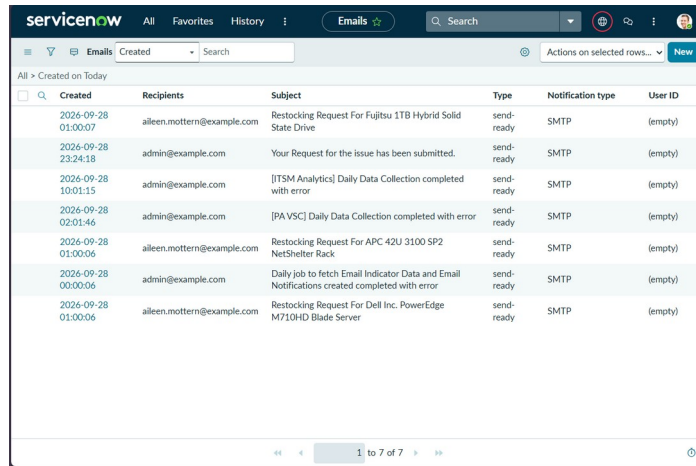
7. RESULTS

7.1 Output Screenshots

The screenshots below document the implemented ServiceNow workflow, classification result, notification, and deployment package.

The screenshot shows a ServiceNow record for a "School IT Ticket" with the number "TKT0001012". The record is titled "School IT Ticket - TKT0001012". The "Number" field is highlighted with a green box. The "Short description" is "Wi-Fi not working in library". The "Category" is "Network" and the "Subcategory" is "Wi-Fi". The "State" is "None". The "Caller" is "System Administrator". The "Assigned Group" and "Assigned to" fields are empty. The "Update" and "Delete" buttons are visible at the bottom.

Figure 7: School IT Ticket showing automatic Category and Subcategory classification.



The screenshot shows the ServiceNow 'Emails' list. The table contains 7 rows of email logs. The first row shows an email from 'aileen.mottorn@example.com' with the subject 'Restocking Request For Fujitsu 1TB Hybrid Solid State Drive'. The second row shows an email from 'admin@example.com' with the subject 'Your Request for the issue has been submitted.' The third row shows an email from 'admin@example.com' with the subject '[ITSM Analytics] Daily Data Collection completed with error'. The fourth row shows an email from 'admin@example.com' with the subject '[PA VSC] Daily Data Collection completed with error'. The fifth row shows an email from 'aileen.mottorn@example.com' with the subject 'Restocking Request For APC 42U 3100 SP2 NetShelter Rack'. The sixth row shows an email from 'admin@example.com' with the subject 'Daily job to fetch Email Indicator Data and Email Notifications created completed with error'. The seventh row shows an email from 'aileen.mottorn@example.com' with the subject 'Restocking Request For Dell Inc. PowerEdge M710HD Blade Server'.

Created	Recipients	Subject	Type	Notification type	User ID
2026-09-28 01:00:07	aileen.mottorn@example.com	Restocking Request For Fujitsu 1TB Hybrid Solid State Drive	send-ready	SMTP	(empty)
2026-09-28 23:24:18	admin@example.com	Your Request for the issue has been submitted.	send-ready	SMTP	(empty)
2026-09-28 10:01:15	admin@example.com	[ITSM Analytics] Daily Data Collection completed with error	send-ready	SMTP	(empty)
2026-09-28 02:01:46	admin@example.com	[PA VSC] Daily Data Collection completed with error	send-ready	SMTP	(empty)
2026-09-28 01:00:06	aileen.mottorn@example.com	Restocking Request For APC 42U 3100 SP2 NetShelter Rack	send-ready	SMTP	(empty)
2026-09-28 00:00:06	admin@example.com	Daily job to fetch Email Indicator Data and Email Notifications created completed with error	send-ready	SMTP	(empty)
2026-09-28 01:00:06	aileen.mottorn@example.com	Restocking Request For Dell Inc. PowerEdge M710HD Blade Server	send-ready	SMTP	(empty)

Figure 9: ServiceNow Email log showing the acknowledgment notification.

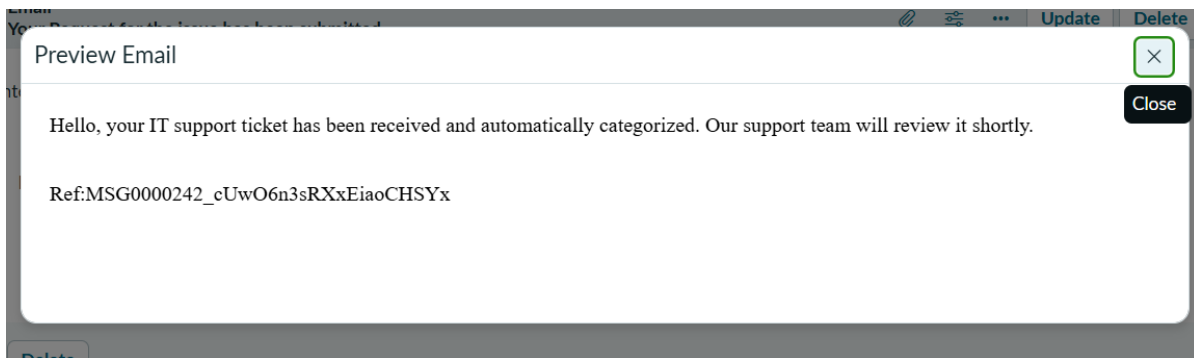
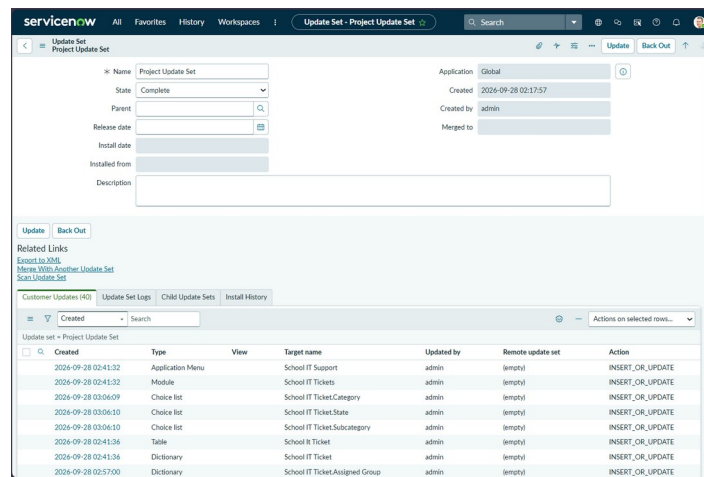


Figure 10: Acknowledgment email preview.



The screenshot shows the ServiceNow 'Update Set - Project Update Set' form. The form includes fields for Name, State, Parent, Release date, Install date, Installed from, and Description. The 'State' is set to 'Complete'. The 'Created' date is 2026-09-28 02:17:57. The 'Created by' is admin. The 'Merged to' field is empty. Below the form, there are 'Update' and 'Back Out' buttons. There are also 'Related Links' for 'Export to XML', 'Merge With Another Update Set', and 'Sync Update Set'. Below the form, there is a table showing the 'Update Set Logs'.

Created	Type	View	Target name	Updated by	Remote update set	Action
2026-09-28 02:41:32	Application Menu		School IT Support	admin	(empty)	INSERT_OR_UPDATE
2026-09-28 02:41:32	Module		School IT Tickets	admin	(empty)	INSERT_OR_UPDATE
2026-09-28 03:06:09	Choice list		School IT Ticket Category	admin	(empty)	INSERT_OR_UPDATE
2026-09-28 03:06:10	Choice list		School IT Ticket State	admin	(empty)	INSERT_OR_UPDATE
2026-09-28 03:06:10	Choice list		School IT Ticket Subcategory	admin	(empty)	INSERT_OR_UPDATE
2026-09-28 02:41:36	Table		School IT Ticket	admin	(empty)	INSERT_OR_UPDATE
2026-09-28 02:41:36	Dictionary		School IT Ticket	admin	(empty)	INSERT_OR_UPDATE
2026-09-28 02:57:00	Dictionary		School IT Ticket Assigned Group	admin	(empty)	INSERT_OR_UPDATE

Figure 11: Completed Project Update Set with customer updates and XML export option.

7.1 Output Screenshots - Configuration Evidence

The screenshot shows the 'Dictionary Entry - Caller' configuration page in ServiceNow. The 'Table' is 'School IT Ticket [u_school_it_tick...]' and the 'Type' is 'Reference'. The 'Column label' is 'Caller' and the 'Column name' is 'u_caller'. The 'Application' is 'Global'. The 'Active' checkbox is checked. The 'Function field' is empty. The 'Mandatory' checkbox is unchecked. The 'Display' checkbox is unchecked. The 'Read only option' is set to '-- None --'. The 'Default Value' section is visible, with a text box for the default value.

Figure 12: Short Description field configuration.

The screenshot shows the 'Dictionary Entry - New record' configuration page in ServiceNow. The 'Table' is 'School IT Ticket [u_school_it_tick...]' and the 'Type' is 'String'. The 'Column label' is 'Short description' and the 'Column name' is 'u_short_description'. The 'Max length' is set to 160. The 'Application' is 'Global'. The 'Active' checkbox is checked. The 'Function field' is empty. The 'Mandatory' checkbox is unchecked. The 'Display' checkbox is unchecked. The 'Read only option' is set to '-- None --'. The 'Default Value' section is visible, with a text box for the default value.

Figure 13: Caller reference field configuration.

The screenshot shows the 'Dictionary Entry - Caller' configuration page in ServiceNow. The 'Table' is 'School IT Ticket [u_school_it_tick...]' and the 'Type' is 'Reference'. The 'Column label' is 'Caller' and the 'Column name' is 'u_caller'. The 'Application' is 'Global'. The 'Active' checkbox is checked. The 'Function field' is empty. The 'Mandatory' checkbox is unchecked. The 'Display' checkbox is unchecked. The 'Read only option' is set to '-- None --'. The 'Default Value' section is visible, with a text box for the default value.

Figure 14: Category field configuration.

Figure 15: Subcategory configured as dependent on Category.

8. ADVANTAGES & DISADVANTAGES

Advantages

- Simple requester experience with basic input fields.
- Automatic classification for supported issue keywords.
- Consistent Category/Subcategory mapping.
- Immediate requester acknowledgment.
- Structured custom ticket storage.
- Controlled change tracking through Update Sets and XML export.

Disadvantages

- Keyword rules depend on how the problem is written.
- Unsupported or vague descriptions may require manual review.
- The current prototype is rule-based rather than model-based.
- No measured performance benchmark is included.
- Automatic technician assignment is not implemented.

9. CONCLUSION

The Auto-Ticket Classification using Flow Designer project demonstrates a complete ServiceNow workflow for simplifying school IT helpdesk ticket processing. The implementation combines a custom data model, dependent choices, event-driven classification, email acknowledgment, access control, and Update Set based deployment tracking.

The supplied screenshots provide functional evidence for ticket creation, automatic classification, notification and finalization. The completed project package can be stored and versioned using the exported Update Set XML.

10. FUTURE SCOPE

- Expand keyword coverage and add fallback handling for ambiguous requests.
- Introduce automatic assignment to support groups.
- Add status-change and resolution notifications.
- Add dashboards for ticket volume, classification and response trends.
- Measure response time and perform controlled load testing.
- Evaluate AI-assisted classification as a future enhancement with appropriate testing and governance.

11. APPENDIX

Source Code (if any)

No separate application source code is required for the implemented prototype. The main implementation is ServiceNow configuration: custom table and fields, choices, Flow Designer logic, email notification, ACLs and Update Set configuration. The completed Update Set was exported as an XML deployment package.

Dataset Link

Not applicable. The implemented workflow does not depend on a machine-learning dataset.

GitHub & Project Demo Link

GitHub Repository: <https://github.com/Akash717656/Auto-Ticket-Classification-using-Flow-Designer>

Project Tracking / Review Board: <https://myskillwallet.ai/dashboard/skillwallet/module/servicenow-system-administrator-nm-eng-6a69e3438beabdd402737035/group-projects/6a96be3b602166829775618a/Auto-Ticket-Classification-using-Flow-Designer-6aba067eca93a2750962af22?tab=3>

Live public demo: Not provided. The application runs in the ServiceNow environment used for the project.